



Freelance Technology Consultancy

JudicialPredict

Predictive Analytics Platform for Judicial Case Outcomes

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Table of Contents

1.	Executive Summary	3
2.	Project Overview & Objectives	3
3.	Scope of Work	4
4.	Data Sources & Predictive Factors	4
5.	Technical Architecture	5
6.	Machine Learning & Statistical Methodology	6
7.	Key Features & User Experience	7
8.	Civil vs Criminal Case Handling	7
9.	Delivery Milestones & Timeline	8
10.	Team Assignments	9
11.	Deliverables	9
12.	Risks & Mitigations	10
13.	Commercial Summary & Next Steps	10

1. Executive Summary

JudicialPredict is a sophisticated predictive analytics and prediction-market platform designed to model and forecast the outcomes of judicial cases and appeals. Drawing on structured court data, historical rulings, party profiles, attorney records, and publicly available information, the system applies ensemble machine-learning models, Bayesian inference, and graph analytics to generate real-time, calibrated probability estimates for case outcomes.

The platform serves legal professionals, researchers, journalists, and market participants who wish to track, analyse, and take positions on judicial proceedings. A live 'ticker tape' bar provides continuous probability updates as each new development - motion, ruling, witness testimony, or expert opinion - is entered into the system. Separate analytical frameworks handle civil and criminal matters, reflecting the fundamentally different standards and dynamics of each.

This document constitutes the full software specification and project plan, including technical architecture, data model, ML methodology, phased delivery milestones, team assignments, and commercial framework.

2. Project Overview & Objectives

Vision

To build the leading intelligence platform for judicial case prediction - empowering legal practitioners and market participants with data-driven probabilistic insights that are transparent, explainable, and continuously updated.

Primary Objectives

- Provide calibrated outcome probabilities for civil and criminal cases and appeals.
- Enable a real-time prediction market where registered users can submit and track forecasts.
- Ingest and process structured data from court records, party databases, and public sources.
- Surface attorney-judge interaction analytics and historical bias indicators.
- Deliver a live ticker tape feed that updates as case events unfold.
- Distinguish meaningfully between civil and criminal proceedings in all models and UX.
- Provide explainable AI outputs - not a black box, but auditable factor breakdowns.

3. Scope of Work

In Scope

- Full-stack web application (desktop & mobile-responsive).
- Data ingestion pipelines for court records, party profiles, and public data.
- ML prediction engine with ensemble models and Bayesian updating.
- Graph analytics layer (attorney-judge relationships, win-rate networks).
- Real-time ticker tape UI component with WebSocket updates.
- Prediction market module - users submit probability estimates and track accuracy.
- Civil case module: damages prediction, liability split, injunction likelihood.
- Criminal case module: conviction probability, sentencing range, acquittal likelihood.
- Admin dashboard for data management, model monitoring, and audit logging.
- REST + WebSocket API for potential third-party integrations.
- Role-based access control: public viewer, registered predictor, legal professional, admin.
- Deployment on Kubernetes with full CI/CD pipeline.

Out of Scope

- Real-money wagering / regulated gambling functionality (architecture will support it in Phase 2).
- Mobile native applications (iOS / Android).
- Automated legal document ingestion via OCR (scheduled for Phase 2).
- Integration with live court e-filing systems.

4. Data Sources & Predictive Factors

Judge Profile Factors

- Historical ruling record by case type, plaintiff/defendant class, and attorney.
- Reversal-on-appeal rate - indicates susceptibility to higher-court override.
- Sentencing leniency/severity index vs jurisdiction mean (criminal cases).
- Temporal bias: ruling patterns by day-of-month, month, court session load.
- Recusal history and disclosed conflicts of interest.

Party & Attorney

- Plaintiff / defendant profiles: prior litigation history, win/loss record, jurisdiction.
- Attorney win rate by case type, court level, opposing counsel, and judge.
- Attorney-judge interaction graph: historical outcomes in prior shared cases.
- Law firm resources, specialist expertise, and seniority of lead counsel.
- Publicly available personal/corporate information about plaintiff and defendant.

Case Dynamics

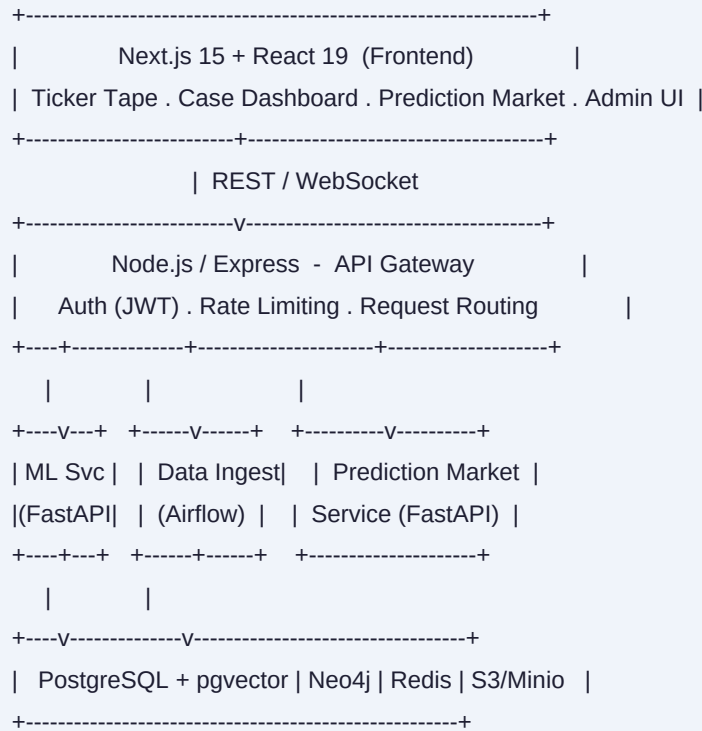
- Procedural history: motions filed, granted/denied ratios, continuances.
- Expert witness credentials, field, and prior testimony outcomes.
- Evidence rulings: admissibility decisions and their historical impact.
- Real-time event feed: new filings, rulings, and hearing outcomes.
- Precedent similarity score to settled landmark cases.
- Civil: financial exposure of defendant, insurance coverage, settlement pressure signals.
- Criminal: charge severity, prosecutorial track record, bail/remand status.

5. Technical Architecture

System Overview

The platform is built as a set of loosely-coupled services communicating over REST and message queues, deployed on Kubernetes. The front end is a Next.js 15 single-page application that connects to an API gateway backed by Python microservices for the prediction engine and data pipelines.

Architecture Diagram (logical)



Technology Stack

- Frontend: Next.js 15, React 19, Tailwind CSS 4
- API Gateway: Node.js 20, Express 5
- ML Service: Python 3.12, FastAPI, Uvicorn
- Data Pipelines: Apache Airflow 2.9
- Realtime: Socket.IO / WebSocket
- Primary DB: PostgreSQL 16 + pgvector
- Graph DB: Neo4j 5 (attorney-judge graph)
- Cache / Queues: Redis 7
- Object Storage: S3 / MinIO
- Orchestration: Kubernetes (K8s), Helm

Key Design Principles

- Event-driven updates: case events published to Redis Streams; ML service subscribes and recomputes probabilities.
- Explainability-first: every prediction includes a SHAP factor breakdown.
- Auditability: all model versions, inputs, and outputs are stored with immutable timestamps.
- Horizontal scalability: stateless ML service instances behind a load balancer.
- Security: JWT authentication, row-level security in PostgreSQL, TLS everywhere.

6. Machine Learning & Statistical Methodology

The prediction engine uses a multi-layer ensemble approach. Base models capture different signal types; a meta-learner combines their outputs into a final calibrated probability.

Base Model Layer

XGBoost / LightGBM Ensemble

Tabular features: judge stats, attorney win rates, procedural counts, temporal variables. Primary workhorse for structured prediction.

Bayesian Network

Captures causal dependencies between factors (e.g. judge x attorney interaction -> ruling). Enables transparent, interpretable inference chains.

Random Forest

Robust baseline; provides feature importance scores used in SHAP explanations.

Legal-BERT NLP Model

Fine-tuned on legal text to encode case descriptions, motion text, and expert testimony into semantic vectors stored in pgvector.

Graph Neural Network (Neo4j + PyG)

Models the attorney-judge relationship graph. Node embeddings capture latent interaction patterns beyond raw win rates.

ARIMA / Prophet (Temporal)

Captures seasonal and longitudinal bias patterns (day-of-week, court calendar effects, judge tenure phase).

Meta-Learner & Calibration

- Stacked generalisation: a logistic meta-learner combines base-model outputs.
 - Platt scaling applied to convert raw scores to well-calibrated probabilities.
 - Brier Score and Expected Calibration Error tracked continuously in model monitoring.
 - Bayesian updating: posterior probabilities updated in real time as case events arrive.
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- Nightly retraining on new case outcomes ingested during the day.
 - Model versioning in MLflow; champion/challenger framework for safe promotions.
 - Drift detection via Population Stability Index on key feature distributions.

Continuous

7. Key Features & User Experience

Live Ticker Tape

A horizontally-scrolling, auto-updating probability bar runs persistently at the top of every case page. It displays the current outcome probability and a directional arrow indicating whether it has moved up or down since the last event. Each new court filing, ruling, or witness testimony triggers a WebSocket push that updates the ticker within seconds.

Case Dashboard

- Summary card: parties, judge, court level, current probability, market consensus.
- Factor breakdown panel: SHAP waterfall chart showing top 10 probability drivers.
- Timeline: chronological event log with associated probability movements.
- Attorney-Judge history graph: interactive network visualisation.
- Expert witness panel: credentials, field, prior testimony outcome record.
- Comparables: top 5 precedent cases by similarity score.

Prediction I

- Registered users submit probability estimates on case outcomes.
- Aggregate market consensus displayed alongside model prediction.
- Accuracy scoring: users ranked by Brier Score over all their predictions.
- Leaderboard and expert badge system for consistently accurate predictors.
- Historical performance graph per user.

Admin & Da

- Case ingestion UI: manual entry and bulk CSV import.
- Model monitoring dashboard: accuracy, calibration, drift metrics.
- Audit log: full immutable history of all data changes.
- Role management: assign/revoke legal professional access.

8. Civil vs Criminal Case Handling

Civil Cases

- Outcome targets: liability (binary), damages quantum (regression), injunction granted (binary).
- Key additional factors: insurance coverage, defendant financial capacity, settlement history.
- Discovery phase signals: number of depositions, document production volume, sanctions motions.
- Expert witnesses weighted by financial/technical domain credibility.
- Appellate module: probability of reversal, partial reversal, or affirmation.

Criminal Ca

- Outcome targets: conviction (binary per charge), sentence range (regression), acquittal.
- Key additional factors: charge severity class, prior criminal record, bail/remand status.
- Prosecutorial stats: charge-to-conviction rate, plea bargain frequency, dismissal rate.
- Jury vs bench trial distinction with separate sub-models.
- Post-conviction: appeal success probability, sentence reduction likelihood.
- Separate visualisation language: conviction probability vs liability probability.

9. Delivery Milestones & Timeline (16 Weeks)

Phase 1 - Foundation (Weeks 1-4)

Infrastructure, data model, core ingestion pipelines, and authentication. Deliverable: running Kubernetes cluster with live database, API gateway, and basic case/judge/attorney data populated from seed dataset.

Phase 2 - Core Product (Weeks 5-9)

ML pipeline v1, ensemble model training, ticker tape component, case dashboard, SHAP explainability layer, and prediction market MVP. Deliverable: internal beta - end-to-end flow from case event to probability update on screen.

Phase 3 - Civil & Criminal Specialisation (Weeks 10-13)

Civil and criminal sub-models, attorney-judge graph analytics, expert witness module, appellate branch. Full Bayesian updating pipeline. Deliverable: feature-complete application entering QA.

Phase 4 - Hardening & Launch (Weeks 14-16)

Performance testing, security audit, model calibration validation, documentation, staging-to-production promotion, and client handover. Deliverable: production deployment with runbook and operator training.

#	Milestone	Start	End	Owner
1	Kubernetes cluster + CI/CD pipeline live	Wk 1	Wk 2	Casey M.
2	PostgreSQL + Neo4j + Redis schema deployed	Wk 1	Wk 3	Chris N.
3	Data ingestion pipelines (Airflow) - seed data	Wk 2	Wk 4	Chris N.
4	API gateway + JWT auth complete	Wk 3	Wk 4	Chris N.
5	ML ensemble v1 trained and serving predictions	Wk 5	Wk 7	Chris N.
6	Ticker tape WebSocket component live	Wk 6	Wk 8	Chris N.
7	Case dashboard + SHAP explainability	Wk 7	Wk 9	Chris N.
8	Prediction market module	Wk 8	Wk 9	Chris N.
9	Civil sub-model + damages regression	Wk 10	Wk 11	Chris N.
10	Criminal sub-model + sentencing regression	Wk 10	Wk 11	Chris N.
11	Attorney-Judge graph neural network	Wk 11	Wk 12	Chris N.
12	Expert witness & appellate modules	Wk 12	Wk 13	Chris N.
13	QA - automated test suite + regression	Wk 13	Wk 14	Riley S.
14	Security audit + penetration test	Wk 14	Wk 15	Casey M.
15	Production deployment + client handover	Wk 15	Wk 16	Alex R.

10. Team Assignments

Name	Role	Responsibility
Alex Reeves	Operations Director	Overall delivery, client communication
Jamie Okafor	Project Manager	Sprint planning, backlog, TDD enforcement
Chris Novak	Lead Engineer / CTO	Architecture, ML pipeline, all backend
Casey Muller	DevOps Engineer	Kubernetes, CI/CD, security, monitoring
Riley Svensson	QA Engineer	Test plans, automated suites, acceptance
Jordan Whitaker	Customer Delivery	Client-side review & approval gate

11. Deliverables

Deliverable	Format	Responsible
Production web application	Deployed URL + access creds	Chris N. / Casey M.
Source code repository	Private GitHub repo	Chris N.
Architecture & data model docs	Markdown + diagrams	Chris N.
ML model registry + runbooks	MLflow + Markdown	Chris N.
Automated test suite	Jest / pytest / Playwright	Riley S.
Test report	PDF / HTML report	Riley S.
CI/CD pipeline + Helm charts	GitHub Actions + K8s YAML	Casey M.
Operator runbook	PDF	Casey M.
User documentation	Web-based docs site	Jamie O.
Project retrospective	Written summary	Alex R.

12. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Data Availability	High	Medium	Court records may be sparse or inconsistently formatted. Mitigation: begin data audit in Week 1; agree minimum viable dataset with client.
Model Accuracy	Medium	Low	Initial models may underperform without sufficient historical data. Mitigation: publish confidence intervals; communicate calibration metrics transparently.
Legal / Ethical	Medium	High	Predictions may be perceived as influencing proceedings. Mitigation: clearly label outputs as statistical estimates; legal counsel review of T&Cs.
Scope Creep	Medium	Medium	Stakeholder requests may expand scope mid-delivery. Mitigation: formal change control process; all additions assessed against appetite budget.
Third-Party Data APIs	Low	Medium	External data sources may change or become unavailable. Mitigation: abstract behind adapter layer; maintain local cache of ingested data.

13. Commercial Summary & Next Steps

Indicative Pricing

Based on the scope defined above, we estimate this engagement at the Premium tier of our AI & Automation practice. A detailed fixed-price proposal will be issued once the following open items are resolved:

- Confirmed geographic scope (Croatian courts only vs multi-jurisdiction).
- Data handover format from the related Croatian Judiciary project.
- Decision on real-money prediction market vs analysis-only in Phase 1.
- Authentication model: open registration, invite-only, or enterprise SSO.
- Language requirements: Croatian, English, or multilingual.
- Target launch date and any hard deadline constraints.
- Budget envelope confirmation.

Immediate Next Steps

- Client reviews this specification and returns annotated feedback within 5 business days.
- GigForge schedules a kick-off call to resolve open items above.
- Data audit commenced in parallel - client to provide sample court record exports.
- Fixed-price proposal and Statement of Work issued following kick-off call.
- Sprint 1 commences upon SOW signature.

Contact

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